

Offline Flyback SMPS Design — Detailed Design Notes & Standards

References

1. Scope and Design Intent

This document provides a comprehensive engineering reference for the design of an offline isolated flyback SMPS. It consolidates schematic, layout, safety, EMI, and Altium rule decisions with references to applicable safety standards.

2. Applicable Safety Standards

The following standards inform spacing, isolation, and component selection:

- IEC 62368-1 (Audio/Video, ICT Equipment Safety)
- IEC 60950-1 (Legacy IT Equipment Safety)
- IEC 60664-1 (Insulation Coordination)
- UL 62368-1

For 120 VAC mains, Pollution Degree 2, Material Group IIIa is assumed unless otherwise stated.

3. Safety Capacitors

X Capacitors (X1/X2):

- Used line-to-line (L-N)
- Suppress differential-mode EMI
- Must be safety certified

Y Capacitors (Y1/Y2):

- Used primary-to-secondary
- Provide common-mode EMI return
- Must maintain reinforced/basic insulation

Ordinary ceramic capacitors are never acceptable for mains or isolation use.

4. Grounding Strategy

Primary and secondary grounds must be split at the schematic level:

- PRI_GND — primary return (hot side)
- SEC_GND — secondary return (cold side)

Grounds are connected only through optocoupler and Y-capacitor. This is required by IEC insulation coordination principles.

5. Line and Neutral Handling

Line (L) and Neutral (N) are treated as live conductors.

They are routed as traces, never poured, and must meet clearance requirements.

6. EMI Input Filtering Strategy

A common-mode choke footprint is retained but marked DNP for Rev-A.

0-ohm jumpers allow bypass during bring-up.

This matches industry practice for early prototypes.

7. Isolation Geometry

Copper keepouts are used across the isolation barrier.

Slots are strongly recommended to increase creepage per IEC 60664-1.

8. Altium Net Classes

PRIMARY Net Class:

- L, N, HV+, PRI_GND, Drain

SECONDARY Net Class:

- SEC_GND, VBUS, 5V, USB D+/D-

9. Clearance Rules

Primary-to-secondary clearance rule:

- Scope: PRIMARY ↔ SECONDARY
- Minimum clearance: 6.0 mm (120 VAC)

Primary internal clearance:

- 1.5–2.0 mm

10. Verification Checklist

- ERC clean
- DRC clean
- Polygons repoured
- Clearance violations = 0
- Only opto/Y-cap cross isolation

11. Design Philosophy

Isolation must be enforced by:

1. Schematic net separation
2. PCB rules
3. Physical geometry

This layered approach is used in professional offline SMPS designs.